CROSS-GENRE SONG TRANSFORMER: INDIVIDUAL PROGRESS REPORT

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ABSTRACT

This report documents my arrangement and experimental contributions to a project transforming MIDI excerpts between Jazz and Rock presets while retaining a selected melody. The prototype adapts chord-relative patterns and selects phrase plans with continuity costs and repeated-pattern constraints. A runner compares instrument substitution, fixed patterns, adaptive selection, and a continuity ablation. A source-derived pilot uses twelve compositions: four for development and eight held out from parameter selection. All 48 exports pass corrected technical checks and an independent raw-MIDI audit; the suite passes 42 tests. Adaptive selection changes four of eight test arrangements. Provisional annotations, an inconclusive audio-model evaluation, and pending listening limit conclusions about perceived genre improvement. Final goals prioritize reviewed inputs and independent evaluation.

1. INTRODUCTION

The team project asks whether a melody-constrained, style-aware arranger can produce a more recognizable target style than instrument substitution while retaining source identity. The core scope is approximately 30–60-second instrumental MIDI excerpts in steady-tempo 4/4, aligned to musical phrases. Melody pitches, onsets and durations, tempo, and supplied harmony and boundaries are protected. Accompaniment instruments, rhythms, bass motion and chord voicings may change. MIDI and playable audio are the intended outputs; audio transcription, vocals, unrestricted genres and large-model training are outside the core scope.

My objectives are implementing the adaptive arranger and measuring how arrangement choices affect outputs. Smit Amin owns corpus preparation and project baselines; Vinay Kumar Devarakonda owns style modelling and evaluation; Abdullah Shah owns rendering and integration. My standalone comparators exercise the runner; teammate components are credited separately. Code and checkpoint evidence are published in the project review branch. [1]

[1] https://github.com/Pyrex111/CSC575_Music_Retrieval_Techniques/pull/1

2. RELATED WORK

Lu and Su separate prominent melody from accompaniment in homophonic style transfer [1]. Symbolic CycleGAN [2] addresses unpaired genre transfer, while MIDI-VAE [3] models dynamics and instrumentation. These approaches motivate changes beyond instrument substitution; this project instead makes melody preservation an explicit constraint. Cifka et al. use synthetic aligned data for supervised translation [4], followed by one-shot accompaniment transfer in Groove2Groove [5]. Their content/style distinction informs the design; a pretrained comparison remains unimplemented.

MusicVAE [6], Music Transformer [7], and Pop Music Transformer [8] motivate longer-term organization and beat alignment. My prototype uses supplied boundaries and a small candidate library rather than reproducing those architectures. The Jazz Transformer analysis [9] and Yang and Lerch's evaluation study [10] support separating structural measurements from listening judgments. MusPy [11] provides a related symbolic-evaluation toolkit. Lakh [12] and the Groove MIDI Dataset [13] remain resources in the final design; this pilot uses Smit's files, and no GMD patterns have been integrated.

The implementation uses `pretty_midi` [14]. The design identifies `music21` [15] for harmonic analysis and `mir_eval` [16] for standard comparisons; current chord estimates and the raw-event audit are custom. Abdullah's FluidSynth pipeline [17] supplies matched audio. A separate CLAP audio–text model [18] was tested as an evaluation diagnostic, with limitations discussed below.

3. METHOD AND IMPLEMENTATION

3.1 Representation and Constrained Adaptation

Each supported input comprises cropped MIDI, an explicit melody-track index, one supplied chord per bar, constant tempo, excerpt length, and labelled boundaries. Notes use beat coordinates. Chord-relative bass and chord templates map into bass pitches 36–60 and chord pitches 48–84. The melody is copied separately so accompaniment operations cannot overwrite it. Invalid pitches, nonpositive durations and events outside the excerpt are rejected. The current contract does not support tempo changes or within-bar chord changes.

Each preset has three authored pattern levels—sparse,

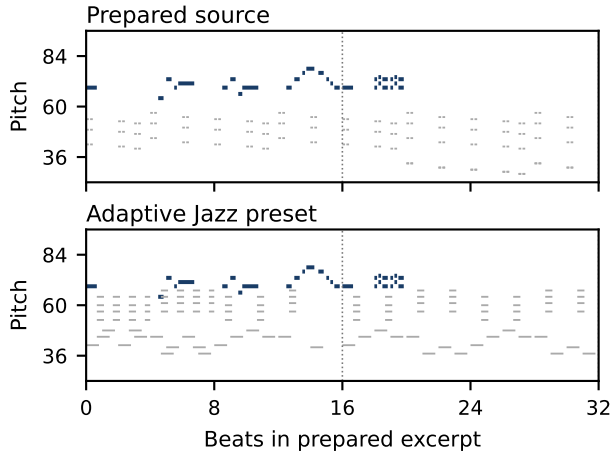


Figure 1. First eight bars of the prepared *Brain Stew* excerpt and adaptive Jazz-preset output. Dark segments show protected melody; light segments show pitched accompaniment. Drums are omitted. Dotted lines mark the provisional four-bar grid.

regular and busy—combined into five plans: three constant levels, build and release. Local cost compares activity with melody density and a preset-specific offbeat profile. These are engineering heuristics. Rhythms and instruments change while supplied harmony remains fixed (Figure 1).

3.2 Phrase-Level Selection

For candidate plans $c_1, \ldots, c_P$, selection minimizes

$$\hat{\mathbf{c}} = \arg\min_{\mathbf{c}\in\mathcal{F}} \alpha \sum_{p=1}^{P} L_p(c_p) + \lambda \sum_{p=2}^{P} T(c_{p-1}, c_p), \qquad (1)$$

where $\mathcal{F}$ enforces supported timing, harmony, register and repeated-label constraints. Repeated labels reuse a plan, adapted to their chords. The transition cost is

$$T(a,b) = \frac{|\ell(a) - f(b)|}{12} + \frac{|\rho(a) - \rho(b)|}{24}, \qquad (2)$$

where ℓ and f are last/first bass pitches and ρ is accompaniment events per bar. Dynamic-programming states retain the preceding candidate and live repeated-label assignments; exceeding the state limit raises an error. Exhaustive search verifies optimization on a small repeated-phrase case.

4. EXPERIMENTAL SETUP

4.1 Inputs and Development Selection

The original checkpoint used four authored synthetic 16-bar fixtures, lasting 32 or 48 seconds, with sixteen method outputs. The source-derived convenience pilot selects twelve distinct compositions from Smit's dataset revision `3316077`. Development cases are *C'est si bon*, *Feels So Good*, *Bad Medicine*, and *Sweet Child O' Mine*; the eight test cases appear in Table 2. Both splits contain both source-folder genres. Each 16-bar crop lasts approximately 30–60 seconds. Composition IDs and exact source/prepared/melody-content hashes do not cross splits; artist independence and alternate-version detection are not established.

Melody tracks were selected using names, instruments and note activity. Chords were estimated from duration/velocity-weighted accompaniment pitch classes with bass support. Four-bar segments share a label only when melody signatures match exactly. These are *provisional computational annotations*, not independently musician-reviewed ground truth. Crop-boundary edits are logged; preservation compares against the prepared excerpt. The estimator flags 173/192 bars under conservative coverage, candidate-margin and half-bar-disagreement rules. Flags indicate uncertainty, not confirmed errors.

I fixed $\alpha = 1$ and compared $\lambda \in \{0, 0.1, 0.25, 0.5, 0.8, 1, 2, 4\}$ on development inputs only. Global rescaling of both weights cannot change the path. The selection proxy combines distance to target-style development-source descriptor profiles with a boundary bass-jump penalty. Vinay's pinned extractor supplies note density, polyphony, offbeat activity and mean duration in beats. Normalization uses development sources only, with recorded scale floors. The mean proxy falls from 1.1599 at $\lambda = 0$ to 1.1190 at $\lambda = 4$, so weights 1/4 were frozen before test evaluation. This is a grid-boundary choice, not a global optimum or perceptual validation.

4.2 Matched Conditions and Reproducibility

B0 retains source accompaniment events and changes pitched accompaniment instruments. B1 uses regular target patterns throughout. M adaptively selects from the same library; the ablation removes continuity. Inputs, melody, annotations, duration and target palettes are matched, and B1/M share the library. The deterministic runner saves IDs, settings, input/code/output hashes, traces, failures and transformation times. No arrangement seed is required. Pinned revisions attribute source data, feature extraction and rendering to their respective teammates.

5. INITIAL RESULTS AND DISCUSSION

5.1 Technical Verification and Diagnostics

All 48 outputs pass corrected technical checks and a separate raw-MIDI audit. The audit reads event streams without importing the generator or its evaluator and finds zero melody pitch/onset/offset/velocity differences in beat coordinates. Tempo, metre, endpoints and supplied markers are retained. Generated B1/M/ablation accompaniment has zero annotated chord-membership, register or event-bound violations. B0's original non-chord tones and registers are descriptive source properties; generator-only constraints do not apply. The suite passes 42 tests, including melody corruption detection and non-first-track selection.

Indicator	Approved task	Current status and evidence
O1 PI1 (B)	Representation and interfaces	Implemented: beat-based contract and worked examples; team annotation review pending.
O1 PI2 (B)	Constrained chord adaptation	Implemented/tested: melody protection, chord mapping and bounds; harmony checks concern supplied annotations.
O1 PI3 (E)	Descriptor retrieval; development parameters	Implemented; eight-setting development pilot completed, weights frozen at 1/4. Profiles/library remain hand-authored.
O1 PI4 (E)	Phrase selection and repeated consistency	Implemented: DP, transitions, repeat tests and both target presets. Real-source boundaries remain provisional.
O1 PI5 (A)	Learned/pretrained matched comparison	Pending: no learned reranker or pretrained accompaniment comparison is claimed.
O2 PI1 (B)	Matched hypothesis and experiment	Implemented: research question, controlled B0/B1/M factors and measured outcomes.
O2 PI2 (B)	Logged experiment runner	Implemented: IDs, settings, hashes, traces, outputs, failures and transformation times.
O2 PI3 (E)	Held-out two-direction comparisons	Pilot completed: eight composition-separated cases, invariants, changes, failures and runtime; diagnostic reuse disclosed.
O2 PI4 (E)	Ablations and interpretation	Symbolic pilot completed with per-composition analysis; specific listener judgments pending.
O2 PI5 (A)	Validation-selected sensitivity	Initial grid completed; broader robustness strata and final-study trade-offs pending.

Table 1. Individual indicator status. O1: adaptive arranger; O2: comparative experiments. B/E/A: basic/expected/advanced. Implementation does not imply validated musical quality or completed final-project evaluation.

Real inputs exposed a discrepancy between BPM floats and encoded integer MIDI tempo, producing false bar-boundary harmony and duration failures. The first pass recorded 36/48 generator-check and 32/48 raw-audit passes. Corrected measurements recheck the identical exports at 48/48 with unchanged hashes. A decimal-tempo serialization fix and three regression cases were added; a fresh corrected-code repeat also passes 48/48 with the same selected weights. Original results, first-run code and corrections are retained. Because test diagnostics were inspected before correction, these are exploratory reused-test results, not a pristine confirmatory hold-out study.

5.2 Comparisons and Ablations

Table 2 shows heterogeneous effects. M differs from B1 in four of eight test compositions; matching B1 elsewhere is a valid outcome. Continuity changes the plan in six cases. Mean phrase texture change is 1.09 notes/bar for full M versus 3.17 without continuity; boundary bass jumps average 1.96 versus 3.08 semitones. However, texture change increases for *Let There Be Love* and *What a Wonderful World*. Smaller measured changes are not necessarily musically preferable.

Target-reference descriptor distance averages 0.792 for B0, 1.073 for B1, 0.976 for M and 1.007 without continuity. M improves on B1 under this proxy; superiority to B0 is not established. Single-run mean transformation times are 2.67, 5.08, 5.54 and 5.38 ms, respectively, on Linux/Python 3.12 CPU, excluding I/O and rendering. These are engineering timings, not a controlled hardware benchmark.

Test composition	ΔE	D_M	D_{-C}
La Vie en rose	0	0.50	0.50
Let There Be Love	280	2.33	1.17
Light Years	50	1.50	8.17
What a Wonderful World	43	1.75	0.83
Brain Stew	132	2.00	8.00
Highway to Hell	0	0.00	0.00
T.N.T.	0	0.50	4.67
When I Come Around	0	0.17	2.00

Table 2. Pilot comparisons. ΔE: accompaniment events differing between M and B1. D: mean adjacent-grid texture change, notes/bar; $-C$: continuity removed. First four rows target Rock; last four target Jazz. These are symbolic measurements, not listener preferences.

5.3 Audio Evaluation Attempt

Abdullah's renderer produced forty matched 44.1-kHz stereo PCM16 WAVs: source plus four conditions per test composition. All are nonzero, without full-scale clipping, and duration-matched within each case. Constant-gain processing targets -23 LUFS with a peak ceiling; measured loudness is -23.05 to -22.98 LUFS.

A pinned quantized CLAP model [18] evaluated waveforms using fixed genre prompts and full-coverage ten-second windows. Its forced Jazz/Rock choice matches only 2/4 development-folder labels; both Rock development renderings favour Jazz. Although target-choice agreement is 5/8 for M and 4/8 for other methods, these are diagnostic counts, not validated accuracy. The control check prevents treating the model as a reliable final judge. Human recognizability/coherence ratings remain uncollected. Ear-

lier informal listening concerned synthetic fixtures only. A reference-guided form is prepared for actual responses.

6. INDIVIDUAL STATUS AND REMAINING WORK

Table 1 maps all ten indicators to evidence. Basic implementation indicators are supported. Expected methods and pilot comparisons are implemented; independently reviewed annotations and perceptual interpretation remain incomplete. The twelve-composition checkpoint is an explicit progress-stage reduction from the original sixty-composition target, not fulfilment of the final corpus plan. The core demonstration runs independently; real-input integration uses credited teammate components.

6.1 Revised Milestones and Final Goals

Basic: retain the CPU arranger, preservation checks, two-direction MIDI outputs and reproducible B0/B1/M comparison as a fallback. This foundation is already demonstrable.

Expected: obtain Smit's annotation/provenance review and agree grouped splits; freeze inputs/settings and rerun comparisons; then use Abdullah's renders and Vinay's independent style/listening analysis. Collect method-anonymized responses before integration and reporting. The original sixty-composition and 8–12-listener targets remain unless the team records a scope revision.

Advanced: integrate a learned reranker or compatible pretrained accompaniment model under the same constraints and compare matched inputs. Extend weight analysis across predefined melody-density and harmony-complexity strata, select on validation data, and report frozen-test trade-offs and uncertainty. A ranker's own score cannot independently establish improvement. These strengthened goals are not counted as completed work.

The standalone arranger supports continued work while team handoffs are arranged. Perceived target-genre improvement remains an open experimental question.

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